

March 9, 2026

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**Re: Reply Comments of the Vehicle-Grid Integration Council on Draft Resolution E-5452. Southern California Edison Orchestrated Charging and Advanced Resiliency for Distribution Vehicle Grid Integration Proposal Using Low Carbon Fuel Standard Holdback Revenue**

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Dear Sir or Madam:

Pursuant to the provisions of General Order 96-B the Comment Letter accompanying Draft Resolution E-5452, the Vehicle-Grid Integration Council (“VGIC”) respectfully replies to comments on the proposed approval with modifications of Southern California Edison’s (“SCE”) Advice Letter 5536-E, *Southern California Edison’s Request for Approval to Update Its Low Carbon Fuel Standard Holdback Revenue Return Implementation Plan to Include a New Vehicle Grid Integration Program and to Qualify for Exemption from Public Utilities Code Section 851* (“Advice Letter”).

## **I. INTRODUCTION.**

VGIC’s Opening Comments on the Draft Resolution highlighted the significant grid, customer, and ratepayer benefits that can be unlocked through bidirectional charging systems and emphasized the importance of approving SCE’s proposed incentives for bidirectional charging systems in its Orchestrated Charging and Advanced Resiliency for Distribution (“ORCHARD”) program.

Opening Comments from other stakeholders demonstrate overwhelming consensus around the value of bidirectional charging systems beyond backup power and the importance of reducing barriers to bidirectional charging system deployment. Additionally, recent analysis from the California Energy Commission (“CEC”), supported by the National Laboratory of the Rockies, which was released after Opening Comments were submitted, strongly supports these same conclusions, highlighting the substantial demand flexibility potential of bidirectional charging systems and identifying upfront equipment and installation costs as a key barrier to deployment. Lastly, new developments in the distribution interconnection proceeding suggest forthcoming delays in reducing the high interconnection application fee, which itself is only one of several major upfront cost barriers to customer adoption.

For these reasons, the Commission should approve SCE’s request to include bidirectional charging incentives within the ORCHARD program and support the continued development of vehicle-grid integration technologies that can provide least-cost flexibility for California’s electric grid.

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## **II. STAKEHOLDERS SHOW BROAD SUPPORT FOR AND GRID VALUE FROM BIDIRECTIONAL CHARGING SYSTEM DEPLOYMENT INCENTIVES.**

### **A. The majority of comments on the Draft Resolution emphasize the importance of the ORCHARD bidirectional incentive proposal.**

A wide range of stakeholders expressed support for authorizing incentives for bidirectional charging systems, including ChargeScape, SCE, WeaveGrid, VGIC, Ford Motor Company, and a Joint Comment Letter signed by 37 organizations. These organizations include Plug In America, Ford Motor Company, American Honda Motor Co., Inc., GM, Tesla, Inc., Toyota, Volvo Cars, Acterra: Action for a Healthy Planet, Advanced Energy United, Alliance for Automotive Innovation, Alliance for Transportation Electrification, Bidirectional Energy, bodeEV, Inc., CALSTART, Ceres, ChargePoint, Inc., ChargeScape, Clean Coalition, Clean Energy Works, Codibly, Inc., Debel LLC, Eaton, Emporia Energy, Energy Hub, Enphase Energy, ev.energy, Fermata Energy, GRID Alternatives, International Energy Agency TCP-EV Task 53, Leap, NEMA, The Climate Center, The Mobility House, Treehouse, VGIC, Wallbox, WeaveGrid, and World Resources Institute.

These comments consistently highlight two key points:

1. **There are valuable use cases for bidirectional charging systems outside of backup power use cases.** This includes providing load reduction and bill management services, with SCE estimating that customers can achieve \$500 in bill savings through EV discharge “offsetting behind-the-meter loads alone,” without exports.<sup>1</sup> ChargeScape also highlights that, “The Draft Resolution’s concerns do not apply to grid-parallel V2H operation where energy flows remain behind the meter and serve only the customer’s own load,” and that **this pathway would be leveraged by customers today if not for the cost barriers currently posed, including high upfront costs** and the \$800 interconnection fee.<sup>2</sup> However, export opportunities are also being pursued, such as SCE’s Vehicle Grid Resource Proposal (VGRP).<sup>3</sup>
2. **Incremental grid value can be provided by bidirectional charging systems:** VGIC’s Opening Comments discussed the significant grid services these systems can provide when operating in parallel with the grid.<sup>4</sup> The Commission is already planning for approximately 13,500 MW of capacity from bidirectional charging systems by 2045,<sup>5</sup> while the CEC estimates that roughly 18.5 GW of vehicle battery storage capacity is already available in

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<sup>1</sup> SCE Opening Comments on Draft Resolution at 2.

<sup>2</sup> ChargeScape Opening Comments on Draft Resolution at 3.

<sup>3</sup> SCE Opening Comments on Draft Resolution at 2.

<sup>4</sup> VGIC Opening Comments at 3-5.

<sup>5</sup> California Public Utilities Commission (“CPUC”), Draft Inputs & Assumptions, 2024 – 2026 Integrated Resource Planning (IRP) at 111.

[https://www.cpuc.ca.gov/-/media/cpuc-website/divisions/energy-division/documents/integrated-resource-plan-and-long-term-procurement-plan-irp-ltpp/2024-2026-irp-cycle-events-and-materials/2025\\_draft\\_inputs\\_and\\_assumptions\\_doc\\_20250220.pdf](https://www.cpuc.ca.gov/-/media/cpuc-website/divisions/energy-division/documents/integrated-resource-plan-and-long-term-procurement-plan-irp-ltpp/2024-2026-irp-cycle-events-and-materials/2025_draft_inputs_and_assumptions_doc_20250220.pdf)

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California today.<sup>6</sup> SCE similarly emphasizes that bidirectional charging incentives can improve system affordability by unlocking flexible capacity from existing EV batteries.<sup>7</sup> As WeaveGrid notes, “Without proper incentives, there may be substantial untapped capacity left on the table that can offer least-cost, best-fit resources for ratepayers.”<sup>8</sup>

**B. A new roadmap from the CEC further shows the benefits of bidirectional charging systems for California.**

On March 5, the CEC released a new *Roadmap to Unlock the Benefits of Bidirectional Charging*, highlighting the significant demand flexibility potential of bidirectional charging systems. The roadmap, developed with support from the National Lab of the Rockies (NLR), finds that by 2035 bidirectional charging systems will be the single largest source of demand flexibility in California.<sup>9</sup> Whether we capture that source to provide value to ratepayers largely rests on the Commission taking decisive action **today** to adopt no-regrets measures to deploy the needed charging infrastructure. Otherwise, customers will be locked into unidirectional charging systems - a stranded asset – and unable to contribute to the grid. The CEC’s vision aligns well with SCE’s proposed LCFS-funded rebate, which would, critically, not rely on already-stressed ratepayer funds, in-demand EPIC research funding, or highly uncertain legislative appropriations.

Importantly, the CEC’s report estimates that **bidirectional discharge can provide approximately three times the value of unidirectional managed charging.**<sup>10</sup> Even without exporting energy to the grid, **residential bidirectional discharge to the home could reduce peak residential demand by up to 5 GW by 2030.**<sup>11</sup>

**At the same time, the CEC identifies high upfront equipment and installation costs as one of the primary barriers to adoption.**<sup>12</sup> Interconnection costs are one important component of these costs; for a \$10,000 system, the \$800 interconnection fee alone represents roughly 8% of total system costs. VGIC provides additional detail below regarding these upfront costs.

Taken together, stakeholder comments and the CEC’s recent analysis demonstrate the substantial value bidirectional charging systems can provide to California’s electric grid and the importance of reducing barriers to deployment. The Commission should therefore approve SCE’s request to provide bidirectional charging incentives through the ORCHARD program.

**C. The interconnection application fee, which may not be reduced expeditiously, is one of many upfront costs that grid-parallel bidirectional charging system adopters must**

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<sup>6</sup> CEC, *A Roadmap to Unlocking the Benefits of Bidirectional Charging* at 2.

<https://efiling.energy.ca.gov/GetDocument.aspx?tn=268952&DocumentContentId=106145>

<sup>7</sup> SCE Opening Comments on Draft Resolution at 2.

<sup>8</sup> Weavegrid Opening Comments at 3

<sup>9</sup> CEC, *A Roadmap to Unlocking the Benefits of Bidirectional Charging* at 4.

<https://efiling.energy.ca.gov/GetDocument.aspx?tn=268952&DocumentContentId=106145>

<sup>10</sup> *Ibid* at 3.

<sup>11</sup> *Ibid* at 1.

<sup>12</sup> *Ibid* at 8.

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**shoulder.**

In opening comments, Ford states, “if there were rebates (or a lower cost interconnection application fee altogether) to help offset the \$800 interconnection application fee for Home Power Management users, it would be able to offer this feature to more customers in California.”<sup>13</sup> Ford further details that “If the Commission is going to deny SCE’s ORCHARD rebates, the high interconnection application fee in California should be quickly taken up as a topic in the Commission’s Rulemaking to Update Distribution Level Interconnection Rules and Regulations (R.25-08-004).”<sup>14</sup> VGIC notes that “the interconnection process and fees for resources that are not interconnection under the Net Billing Tariff (NBT) or Net Energy Metering (NEM) Tariff”<sup>15</sup> is currently slated to be addressed in R.25-08-004, with opening and reply comments on the issue due April 30, 2026 and May 29, 2026, respectively.<sup>16</sup> However, the \$800 interconnection application fee is unlikely to be addressed promptly. The previous interconnection proceeding did not resolve the first bidirectional charging-related issues, which were initially scoped into the proceeding in 2017, until 2020. Notably, the first action taken by the Commission in R.25-08-004 after issuing its March 3, 2026, Scoping Memo and Ruling was to delay comments, which suggests similar delays may be possible throughout this proceeding. Meanwhile, customers **today** seek to adopt bidirectional charging systems to reduce their own energy burden and, in doing so, reduce overall grid costs. Auto OEMs, including Ford and many other Joint Commenters, as well as charging equipment manufacturers, are already offering these systems to certain customers and are eager to expand the offering.<sup>17</sup>

Moreover, as detailed in our Opening Comments, the \$800 interconnection application fee is only one of several categories of upfront cost barriers that grid-parallel bidirectional charging customers currently face. If the \$800 interconnection application fee is reduced before the launch of ORCHARD, several key upfront costs will certainly still remain. Although VGIC anticipates many upfront costs of bidirectional charging system deployment to decline over time as greater

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<sup>13</sup> Ford Motor Company Opening Comments on Draft Resolution at 5.

<sup>14</sup> Ford Motor Company Opening Comments on Draft Resolution at 5.

<sup>15</sup> *Assigned Commissioner’s Scoping Memo and Ruling* in R.25-08-004. Issued March 3, 2026.

<https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M601/K158/601158095.PDF>

<sup>16</sup> R.25-08-004. *Email Ruling Granting Extension for Comments, Motion to Late File NOI, and Motions for Party Status*. Issued March 6, 2026, by ALJ Dugowson.

<sup>17</sup> See Joint Opening Comments submitted by Plug In America, Ford Motor Company, American Honda Motor Co., Inc., GM, Tesla, Inc., Toyota, Volvo Cars, Acterra: Action for a Healthy Planet, Advanced Energy United, Alliance for Automotive Innovation, Alliance for Transportation Electrification, Bidirectional Energy, bodeEV, Inc., CALSTART, Ceres, ChargePoint, Inc., ChargeScape, Clean Coalition, Clean Energy Works, Codibly, Inc., Debel LLC, Eaton, Emporia Energy, Energy Hub, Enphase Energy, ev.energy, Fermata Energy, GRID Alternatives, International Energy Agency TCP-EV Task 53, Leap, NEMA, The Climate Center, The Mobility House, Treehouse, VGIC, Wallbox, WeaveGrid, and World Resources Institute.

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deployment leads to lessons learned, economies of scale, and matured policy frameworks, today's list of upfront costs for a given customer may include:

- **Net generation output meters (NGOM)** for customers on NEM Multiple Tariff (NEM-MT), needing to parse EV discharge from rooftop solar generation for certain customers
- **Site design and engineering costs**, which may be greater for bidirectional systems due to the relatively nascent state of deployment and resulting inefficiencies
- **Certified, bidirectional inverters (if external to the vehicle)**, which are an added component and cost relative to unidirectional charging systems
- **Charging dispensers and/or wall boxes**, including those with embedded inverters, themselves may be higher than unidirectional charging equipment, which is likely due to limited economies of scale in manufacturing/production
- **Gateways, site EMS, and CT coil/clips** to coordinate EV discharge (e.g., to support grid-parallel, *non-exporting* use cases), which may not be needed for unidirectional charging sites
- **Panel and wiring upgrade costs**, which may be greater for bidirectional charging systems than unidirectional charging equipment due to the higher-powered equipment, including for DC-based systems
- **Mounting and safety equipment costs**, which can be greater for bidirectional charging systems than unidirectional charging systems due to the larger size, including for DC-based systems
- **Labor, permitting and inspection, and related installation costs**, which may be greater for bidirectional charging systems due to the relatively nascent state of deployment and resulting lack of streamlining

With these additional upfront cost categories in mind, the likelihood of distribution interconnection proceeding delays, the overwhelming consensus displayed in stakeholders' Opening Comments, the addition of the CEC's newest study to the nine independent studies detailed in VGIC's Opening Comments, the current and forthcoming non-backup-power use cases and products, and the precedent established in the states exhibiting leadership on bidirectional charging, VGIC strongly reiterates its recommendation that the Commission modify the Draft Resolution to approve SCE's request to provide bidirectional charging incentives through the (non-ratepayer-funded) ORCHARD program.

### **III. CONCLUSION.**

VGIC appreciates the opportunity to submit these Reply Comments on the Draft Resolution. We look forward to further collaboration with the Commission and stakeholders on this initiative.

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Respectfully submitted,

/s/ Zach Woogen

Zach Woogen

Executive Director

**VEHICLE-GRID INTEGRATION COUNCIL**